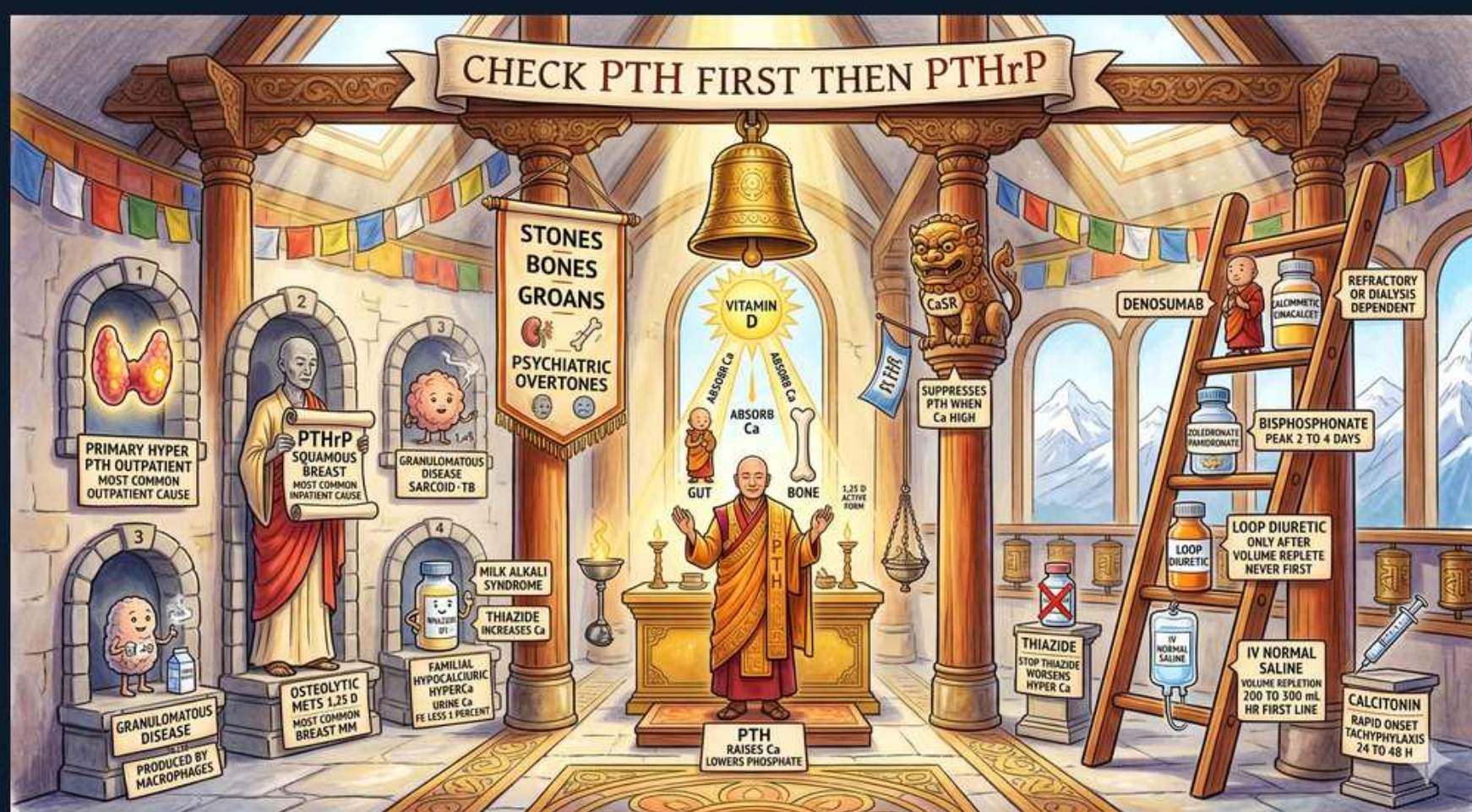


Hypercalcemia Workup — Check PTH First, Then PTHrP



1. Stones, bones, groans, psychiatric overtones

WHAT YOU SEE

Banner of symptom clusters

WHAT IT ENCODES

Classic hypercalcemia symptoms: nephrolithiasis (stones), bone pain/osteitis (bones), abdominal pain/constipation/pancreatitis (groans), and confusion/depression (psychiatric overtones). Severity tracks with the rate and degree of Ca rise.

BOARD TRAP

Mild chronic hypercalcemia (primary hyperPTH) is often asymptomatic; acute marked rises (malignancy) are far more symptomatic — tempo matters.

2. Primary hyperparathyroidism

WHAT YOU SEE

Parathyroid adenoma figure (1)

WHAT IT ENCODES

Primary hyperparathyroidism is the MOST COMMON cause of OUTPATIENT hypercalcemia, usually a single parathyroid adenoma. Labs: high Ca, HIGH/inappropriately normal PTH, low phosphate, high urine Ca. Cure is parathyroidectomy.

BOARD TRAP

High Ca with non-suppressed PTH = PTH-dependent; the next move is to separate primary hyperPTH from FHH using urine calcium, not jump to imaging.

3. PTHrP — squamous/breast malignancy

WHAT YOU SEE

Tumor figure secreting PTHrP (2)

WHAT IT ENCODES

When PTH is SUPPRESSED, check PTHrP. Squamous (lung, head/neck), breast, and renal cancers secrete PTHrP causing humoral hypercalcemia of malignancy with low PTH, low phosphate, low 1,25-vitD.

BOARD TRAP

PTHrP is the most common malignant mechanism — but granulomatous/vitamin-D mediated disease also gives low PTH, so check 1,25-vitD too.

4. Granulomatous disease — sarcoid/TB

WHAT YOU SEE

Granuloma figure (3)

WHAT IT ENCODES

Sarcoidosis and TB make active 1,25-vitamin D via macrophage 1-alpha-hydroxylase, causing low-PTH hypercalcemia with HIGH 1,25-vitD. Steroids suppress it.

BOARD TRAP

Vitamin-D mediated hypercalcemia responds to glucocorticoids, unlike PTH-driven disease — a tested treatment discriminator.

5. Vitamin D — absorb Ca (gut & bone)

WHAT YOU SEE

Central altar: vitamin D over gut/bone

WHAT IT ENCODES

Active 1,25-vitD increases intestinal Ca absorption and mobilizes bone Ca. Exogenous vitamin D toxicity or granulomatous overproduction raises Ca with suppressed PTH; measure both 25- and 1,25-vitD.

BOARD TRAP

25-vitD toxicity (supplements) vs 1,25-vitD excess (granuloma/lymphoma) — the two assays separate the cause.

6. PTH raises Ca, lowers phosphate

WHAT YOU SEE

PTH figure at altar base

WHAT IT ENCODES

PTH raises serum Ca (bone resorption, distal renal reabsorption, renal 1-alpha-hydroxylase) and LOWERS phosphate by blocking proximal reabsorption. This low-phosphate signature is shared by PTH and PTHrP.

BOARD TRAP

Both primary hyperPTH and PTHrP-malignancy give high Ca + LOW phosphate; only PTH level (high vs suppressed) tells them apart.

7. CaSR — suppresses PTH when Ca high

WHAT YOU SEE

CaSR receptor figure

WHAT IT ENCODES

The calcium-sensing receptor on parathyroid cells senses high Ca and shuts off PTH. Calcimimetics (cinacalcet) activate CaSR to lower PTH/Ca in hyperparathyroidism and parathyroid carcinoma.

BOARD TRAP

Inactivating CaSR (FHH) blunts this feedback so PTH stays mildly high — distinguishing it from a true adenoma requires urine Ca.

8. Denosumab

WHAT YOU SEE

Denosumab vial on shelf

WHAT IT ENCODES

Denosumab is an anti-RANKL antibody that blocks osteoclast resorption; useful for malignant hypercalcemia REFRACTORY to or in patients who can't get bisphosphonates (e.g., renal failure, dialysis-dependent).

BOARD TRAP

Unlike bisphosphonates, denosumab is not renally cleared — it's the go-to for bisphosphonate-refractory or dialysis-dependent hypercalcemia.

9. Bisphosphonate — peak 2–4 days

WHAT YOU SEE

Bisphosphonate ladder/vial

WHAT IT ENCODES

IV bisphosphonates (zoledronate, pamidronate) inhibit osteoclasts; onset is DELAYED with maximal effect at 2–4 days, so they are given alongside immediate measures (saline) for malignant hypercalcemia.

BOARD TRAP

Bisphosphonates don't work fast enough alone for symptomatic hypercalcemia — they are paired with IV saline which acts immediately.

10. IV normal saline first

WHAT YOU SEE

Saline bag (treatment ladder)

WHAT IT ENCODES

Aggressive isotonic saline is FIRST-LINE for symptomatic hypercalcemia: it restores volume and promotes calciuresis. Only after the patient is volume-replete should adjuncts be added.

BOARD TRAP

Saline before loop diuretic — giving furosemide to a volume-depleted patient worsens dehydration; volume-replete first.

11. Loop diuretic — only if volume replete

WHAT YOU SEE

Loop diuretic vial with caution

WHAT IT ENCODES

Loop diuretics promote calciuresis but ONLY after volume repletion; routine use is no longer recommended and is reserved for volume overload (e.g., heart/renal failure) during rehydration.

BOARD TRAP

Furosemide is NOT first-line for hypercalcemia anymore — never give it before saline; the old 'saline + loop' teaching has been narrowed.

12. Thiazide — worsens (stop it)

WHAT YOU SEE

Thiazide tablets crossed out

WHAT IT ENCODES

Thiazides reduce urinary Ca excretion and can cause or worsen hypercalcemia; they must be stopped during evaluation/treatment.

BOARD TRAP

A thiazide can unmask mild primary hyperPTH — stopping the drug and rechecking Ca can avoid an unnecessary endocrine work-up.

13. Calcitonin — rapid onset, tachyphylaxis

WHAT YOU SEE

Calcitonin vial

WHAT IT ENCODES

Calcitonin lowers Ca within hours (fast bridge while bisphosphonates take effect) but loses efficacy after ~48 hours due to tachyphylaxis.

BOARD TRAP

Calcitonin's speed is its value, but tachyphylaxis limits it to a 24–48h bridge — it is never definitive therapy.

14. Familial hypocalciuric hypercalcemia (FHH)

WHAT YOU SEE

Familial Ca figure

WHAT IT ENCODES

FHH (inactivating CaSR mutation) gives lifelong mild hypercalcemia, mild high PTH, and LOW urine calcium (Ca/Cr clearance ratio <0.01). It is benign and surgery is harmful/ineffective.

BOARD TRAP

FHH is the great mimic of primary hyperPTH; the 24-hour urine calcium is the single discriminator that prevents needless parathyroidectomy.
